auction information on order routing decisions is uncertain and likely will vary based on market conditions.

The Commission believes that the value of dedicated proprietary auction feeds will be substantially reduced as a result of the addition of auction information to core data, and that this will result in a loss of revenue for those exchanges who offer such feeds.[2008] The Commission believes that the value of any existing data product that provides only auction data[2009] that is not currently in the exclusive SIP feeds will be substantially reduced because of the loss of revenue from these dedicated auction feeds. The Commission expects that many market participants who are executing a trade, either for themselves or for a client, have, and will continue to have, a subscription to core data.

(iv) General Costs to Expanding Consolidated Data

The Commission believes that there are four potential costs to adding the new core data elements, which are common across all these elements. The first potential cost is the cost to the new competing consolidators that will be necessary to implement or upgrade existing infrastructure and software in order to handle the dissemination of the additional core data message traffic. The second potential cost is the cost to SROs to implement system changes required in order to make regulatory data and other data needed to generate consolidated market data available to competing consolidators. The third cost is the technological investments market participants might have to make in order to receive the new core data message traffic. The fourth cost is the cost to users of certain kinds of trading strategies that may currently be relying on the fact that this data is not widely distributed today.

2008 See infra Section V.C.4(a) (discussing effects on exchange proprietary revenue).

2009 See Proposing Release, 85 FR at n.335.

The Commission believes that the cost for firms that wish to become competing consolidators to implement or upgrade infrastructure to handle the dissemination of odd-lot quotes, depth of book information, and auction information will be limited. Competing consolidators will not be required to disseminate all of the information in consolidated market data, including the additional data elements contained in expanded core data, so they will not incur these costs unless they choose to disseminate this information.[2010] As discussed in more detail below,[2011] the Commission believes that the new competing consolidators will likely be firms that already have the technological infrastructure necessary to process full depth of book data and to generate the NBBO using this data. Therefore, for these firms, processing the new message traffic resulting from the additional content of expanded core data may add only a minimal cost to becoming a competing consolidator. However, for a firm that does not currently subscribe to, or process data from, exchange proprietary feeds, the additional message volume will increase the cost of becoming a competing consolidator if they choose to offer a consolidated market data product that includes the additional data elements contained in expanded core data. In particular, if the existing exclusive SIPs should decide to enter the competing consolidator business and choose to offer a consolidated market data product containing this data, they may incur such costs as they do not currently disseminate full depth of

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2010 See supra Section III.C.8(a)(ii).

2011 See infra Section V.C.2(a) for a discussion of the technological capabilities of firms the Commission believes are most likely to become competing consolidators. It is possible that the new definition of core data will make consolidation more difficult for core data than it is currently, and that this added difficulty will result in additional latency. However, the Commission believes that the risk of this is minimal, again because of the technological capabilities of competing consolidators and the market forces that will be in effect in the decentralized consolidation model.

book data. These costs are included in the estimated costs for competing consolidators discussed below in Section V.C.2(d)(i).

The Commission believes that there will be some infrastructure investment required on the part of SROs to provide the information necessary to process and disseminate consolidated market data. The Commission believes that the infrastructure investment required by most SROs to provide the elements necessary to generate core data will be limited, because most SROs currently provide all elements of the new definition of core data over their proprietary feed infrastructure.[2012] In addition, the Commission believes that many competing consolidators and self-aggregators will be firms that already subscribe to these feeds,[2013] and thus, the SROs will likely not have a large amount of new data connections to service and therefore will not need to invest in infrastructure to handle them. However, as discussed by a commenter, FINRA may incur higher infrastructure investment costs in order to make data from the ADF available to competing consolidators and self-aggregators because it currently only provides this data to the SIPs.[2014] Additionally, exchanges, particularly primary markets, may incur some infrastructure costs related to the dissemination of new regulatory data.[2015] Currently, the new regulatory data component to consolidated market data is distributed through the SIPs. In order for this information to be distributed through the new decentralized consolidation model, the rule requires the exchanges to provide a feed to competing consolidators and self-aggregators that contains the regulatory data. The Commission believes that the infrastructure and operational

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2012 See supra Section V.B.2(a).

2013 See infra Sections V.C.2(a) and V.C.2(f).

2014 See FINRA Letter at 3–4. See supra Section III.B.9(e).

2015 As discussed above, this new regulatory data will consist of all the same messages as current regulatory data distributed through the exclusive SIPs. See supra Sections II.H and II.I. See also Proposing Release, 85 FR at Section III.D.

processes to provide such a feed are currently not completely in place and will require investment on the part of exchanges. These costs are included in the estimated costs for SROs discussed below in Section V.C.2(d)(ii).

One commenter stated that requiring each SRO to connect and transmit data to a large number of competing consolidators and self-aggregators could significantly increase costs for SROs.[2016] The Commission disagrees with this commenter. As discussed above, the Commission does not believe that SROs will need to add significant connectivity to account for competing consolidators and self-aggregators, because the Commission believes that most market participants who will become competing consolidators and self-aggregators already subscribe to exchange proprietary data feeds.[2017]

The Commission believes that there will be costs for infrastructure investment in order for market participants to receive the new odd-lot, DOB, and auction information components of core data. However, because market participants will not be required to receive the additional information in core data, the infrastructure investment costs will be limited to those market participants that choose to receive it.[2018] Adding these components to core data will substantially

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2016 See FINRA Letter at 3.

2017 As discussed below, an SRO would incur costs, which could include costs related to expanding connectivity and making sure the data is delivered at similar speeds to its other proprietary feeds, if it developed a separate feed to distribute the data necessary to generate consolidate market data. However the Commission does not believe that an SRO is likely to develop a separate feed and incur the costs. See infra Section V.C.2(d)(v).

2018 See supra Section III.B.6.

increase the total message traffic in core data,[2019] and this increase in message traffic will be accompanied by costs to market participants to set up the infrastructure required to handle this new level of traffic. Commenters stated that this will require significant infrastructure upgrades to receive the data.[2020] The Commission acknowledges that some market participants will require significant infrastructure upgrades to receive the additional elements of core data. However, the Commission notes that the final amendments will not require market participants to receive (or display) the complete set of consolidated market data, and competing consolidators will not be required to deliver all proposed consolidated market data for each data product they offer.[2021] Therefore, most market participants who do not want to incur the costs associated with the expanded core data message traffic due to additional odd-lot information, depth of book information, or auction information will be able to choose not to receive any such additional information. Thus, market participants who do not wish to incur the cost of the infrastructure investments necessary to receive the new core data will not. For those market participants who do wish to incur the cost, the Commission is unable to estimate the associated costs because the

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2019 The Commission believes that the addition of information on odd-lots quotes that are priced at or more aggressively than the NBBO and the addition of DOB information, in particular, may substantially increase message traffic. See Proposing Release, 85 FR at n.294. Commenters agreed that the expansion of new core data, especially the inclusion of DOB would significantly increase message traffic. See, e.g., Virtu Letter at 5; STANY Letter II at 3.

2020 See, e.g., Virtu Letter at 5; TD Ameritrade Letter at 5.

2021 A market participant that has obligations under Rule 603(c) will have to receive all data necessary to generate consolidated market data to comply with the rule. The specific cost associated with some of this data is discussed below. See infra Section V.C.2(d).

costs would vary across market participants and depend on each market participant's existing infrastructure.[2022]

Some commenters stated that the increase in message traffic from expanding core data will increase the latency of core data.[2023] The Commission does not believe that expanding the content of core data will increase the latency of core data when it is combined with the decentralized consolidation model. The Commission believes that competing consolidators will develop technology to handle the expanded content of core data and to reduce the latency of aggregating and transmitting core data.[2024] The Commission understands that third party market data aggregators aggregate and disseminate proprietary DOB feeds (which contain additional message traffic) at lower latencies than the exclusive SIPs and expects that competing consolidators would use similar technology to aggregate and disseminate the expanded content of core data at lower latencies than the exclusive SIPs.[2025] Furthermore, the decentralized consolidation model will also reduce geographical latency by eliminating the extra hop that the exclusive SIPs currently experience.[2026] As discussed above, market participants may also need to expand their bandwidth and invest in additional technology and infrastructure to handle receiving the additional content in core data. The increase in message traffic could increase the latency of market participants receiving expanded core data if they do not make these investments. However, the Commission believes that for those market participants who choose

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2022 See infra note 2290 and accompanying text.

2023 See, e.g., STANY Letter II at 3.

2024 See infra Sections V.C.2(c)(ii) and V.C.2(c)(iii).

2025 See supra Section V.B.2(b).

2026 See infra Section V.C.2(c)(iii).

to receive the entire content of consolidated market data, these market participants will make the investments in technology to receive the data and not add latency.

The Commission believes that adding the odd-lot quote, depth of book, and auction information to core data may impose a cost on traders who rely on strategies that take advantage of the fact that the information in odd-lot quote, depth of book, and auction data is not widely distributed (i.e., those traders who are beneficiaries of existing informational asymmetries). To the extent that some of the value of odd-lot quote, depth of book, and auction information lies in the fact that they currently are not observed by a number of market participants, the Commission believes that the dissemination of this data will adversely impact the profitability of such trading strategies. For traders using trading strategies based on depth of book information, the magnitude of the cost caused by the proposed amendments will depend on the extent to which the five aggregated levels of depth approximate the information contained in the full depth of book information. To the extent that these strategies exploit the lack of information on the part of exclusive SIP-reliant traders, this cost will represent a partial transfer to traders who currently rely solely on SIP data. The Commission is unable to estimate the size of this effect, since it does not have a method for detecting the use of such trading strategies from market data or determining what the profit on such strategies would be if they could be detected.

One commenter stated that the Commission did not evaluate the effects of the potential changes in these trading strategies, including its effects on liquidity on "lit" markets.[2027] The Commission does not believe that changes in these trading strategies will have a significant effect on the liquidity on exchanges because increased competition from new market makers and broker-dealers that receive the expanded content of core data (which contribute to the reduction

[2027] See Nasdaq Letter IV at 31.

in the profits of those traders who are beneficiaries of existing informational asymmetries) will offset any liquidity reduction that may have occurred from changes in the trading strategies of those traders who are beneficiaries of existing informational asymmetries.[2028] However, the Commission is unable to estimate the size of this effect because it cannot estimate the extent to which the profitability of such trading strategies will be affected.

One commenter stated that one cost the Commission did not consider in the expansion of core data was that, to the extent that the definition of core data continues to be updated in the future Commission rulemaking to include more proprietary data in it, exchanges will have less incentive to innovate and provide new or improved proprietary data products.[2029] The Commission agrees that to the extent this happens, the incentive to innovate will be reduced. However, the Commission does not believe that the incentive to innovate will be entirely removed. The final rules do not include all proprietary data elements in consolidated market data and do not contemplate any updates to core data (except for additions to auction data information). Therefore, exchanges may be able to expect that some amount of revenue could be collected on new proprietary data products developed. To the extent that the Commission does not change the definition of core data in the future to include any new data products after such products are made available, the exchanges may be able to collect a significant amount of revenue on such products and therefore will continue to have strong incentives to innovate. To the extent that the Commission frequently changes the definition of core data to include new products developed by exchanges soon after they are made available, exchanges may not be able

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2028 See infra Section V.C.4(a).

2029 See Nasdaq Letter IV at 7 ("Expropriating the proprietary market data products that Nasdaq and others have spent years developing would rob them of the fruits of their labors and dash their incentives to develop new and innovative data products going forward.").

to collect significant revenue from them and their incentives to innovate will weaken. In the event that exchange incentives to innovate are weak, the lost innovation may represent a significant cost to the market.

Commenters stated that retail investors currently receive core data at little or no cost and that the expansion of core data content would increase costs for retail investors.[2030] One of these commenters stated that currently retail investors who do not use depth-of-book data and auction data do not pay for it, but that the proposed rule will replace this with a single feed that is too much data for the retail investor.[2031] The Commission believes that there is uncertainty regarding the cost of market data that retail investors will pay.[2032] One factor would be the data content that retail investors receive. If retail brokers supply retail investors with some of the additional content from expanded core data, then their costs could increase but still be lower than the current cost of receiving equivalent data from SIP and proprietary feeds. However, even if retail brokers do not supply retail investors with any additional content from expanded core data, there are reasons that the overall cost of market data for retail investors could stay at similar rates or decrease relative to the fees charged by the current exclusive SIPs, including, among other

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2030 For commenters' views regarding current retail core data costs see, e.g., Angel Letter at 11 (stating retail "nonprofessional" investors pay almost nothing in direct fees for market data and that most of the data costs are picked up by "professional" users as a result of the good price discrimination in the current system that favors retail investors); Nasdaq Letter IV at 38. For commenters' views regarding cost increases to retail investors from the expansion of core data content, see, e.g., Nasdaq Letter IV at 38; TD Ameritrade Letter at 2, 14–15; Angel Letter at 24.

2031 See Nasdaq Letter IV at 38.

2032 Retail investors may not directly pay for market data, but the costs of retail investors accessing market data may be indirectly passed on through the fees charged by retail broker-dealers.

things, the fees set by the Equity Data Plan(s) and whether they establish fees for data content underlying consolidated market data offerings that use subsets of consolidated market data (i.e., for only TOB data, DOB data, etc.), as well as the different products offered by competing consolidators and how they allocate fixed costs.[2033]

2. Decentralized Consolidation Model

This section focuses on the economic effects pertaining to the decentralized consolidation model. We first discuss the relevant broad economic considerations and economic benefits and costs of the decentralized consolidation model with regards to competing consolidators, then we address economic benefits and costs for self-aggregators, and finally we conclude with the discussion of conforming changes.

(a) Broad Economic Considerations about the Decentralized Consolidation Model

The economic analysis of the effects of the decentralized consolidation model assumes that upon the introduction of the model, a sufficient number of competing consolidators will enter the market so that competitive market forces will have a significant effect on their behavior. Several factors affect the reasonableness of this assumption: barriers to entry into the competing consolidator space, fees for data content, uncertainty regarding connectivity charges for data underlying consolidated market data, potential size of the market for consolidated market data products, and competing consolidators' ability to offer differentiated products. While the Commission recognizes uncertainty in these factors[2034] and that certain economic impacts depend on this assumption, the Commission believes that the risk of too few competing consolidators entering the market, and thus, precluding any potential benefits from materializing

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2033 See infra Section V.C.2(b) for a detailed discussion of these fees.

2034 Commenters agreed that there is uncertainty about the potential market for competing consolidators. See, e.g., Nasdaq Letter IV at 8.